TIRZEPATIDE ONCE WEEKLY FOR THE TREATMENT OF OBESITY

SURMOUNT-1 Trial

Jastreboff AM, Aronne LJ, et al.

Published in *The New England Journal of Medicine,* Jul 2022

Phase 3 Multicenter, Double-blind, Randomized, Placebo-controlled Trial

BACKGROUND & STUDY RATIONALE

About 650 million adults face obesity issues globally, with cardiovascular disease and type 2 diabetes being major contributors, along with a global health burden.

Traditional lifestyle-based approaches have limited effectiveness due to physiological counter-regulatory mechanisms, that hinder sustained weight loss.

Tirzepatide: A Novel Dual Agonist

Glucose-dependent Insulinotropic Polypeptide (GIP) + Glucagon-like Peptide-1 (GLP-1) receptors agonist

Targets: Brain and adipose tissue signaling pathways
Mechanism: Regulates energy balance through dual hormone pathways

Tirzepatide is a once-weekly subcutaneous injectable peptide approved by the U.S. Food and Drug Administration for type 2 diabetes

Study Objective: To evaluated the efficacy and safety of Tirzepatide (5 mg, 10 mg, or 15 mg) compared with Placebo in obesity or overweight adults, excluding those with diabetes.

STUDY DESIGN & METHODOLOGY

Study Population

N = 2,539 adults

Randomized in a **1:1:1:1** ratio to receive **Tirzepatide 5 mg, 10 mg, 15 mg or Placebo**

Treatment Duration

72-week treatment period, included a dose-escalation period of up to 20 weeks

Administration

Subcutaneous injection, once weekly for 72 weeks

Dose Escalation

- ❑ Initiated at a dose of 2.5 mg once weekly (or matching placebo)
- ❑ Increased by 2.5 mg every 4 weeks
- ❑ Reached a maintenance dose of up to 15 mg once weekly by week 20

Inclusion Criteria

- ✓ Adults ≥ 18 years of age
- ✓ BMI ≥ 30 kg/m^2 or ≥ 27 kg/m^2, with at least one weight related complication (e.g. hypertension, dyslipidemia, obstructive sleep apnea or cardiovascular disease)
- ✓ Reported ≥ 1 unsuccessful dietary weight loss attempt
- ✓ Provided written informed consent

Exclusion Criteria

- × Adults with diabetes
- × > 5 kg body-weight change within 90 days prior to screening
- × Previous or planned bariatric surgery
- × Use of weight loss medications within 90 days prior to screening

All participants received lifestyle intervention consisting of dietary counseling (500 kcal/day deficit) and at least 150 minutes/week of physical activity.

STUDY ENDPOINTS & ASSESSMENT

Coprimary Endpoints

- Percentage change in body weight from baseline to Week 72
- Weight reduction of ≥ 5% at Week 72

Secondary Endpoints

- Weight reduction of ≥ 10%, ≥ 15%, and ≥ 20% at Week 72
- Change in weight from baseline to week 20
- Change in waist circumference, systolic BP, fasting insulin, and lipids from baseline to week 72
- Change in QoL: 36-Item Short Form Health Survey (SF-36) physical function score from baseline to week 72
- Percentage change in total body-fat mass (subgroup, n=255, DXA) from baseline to week 72

Safety Assessment: Adverse Events, Serious Adverse Events and laboratory assessments

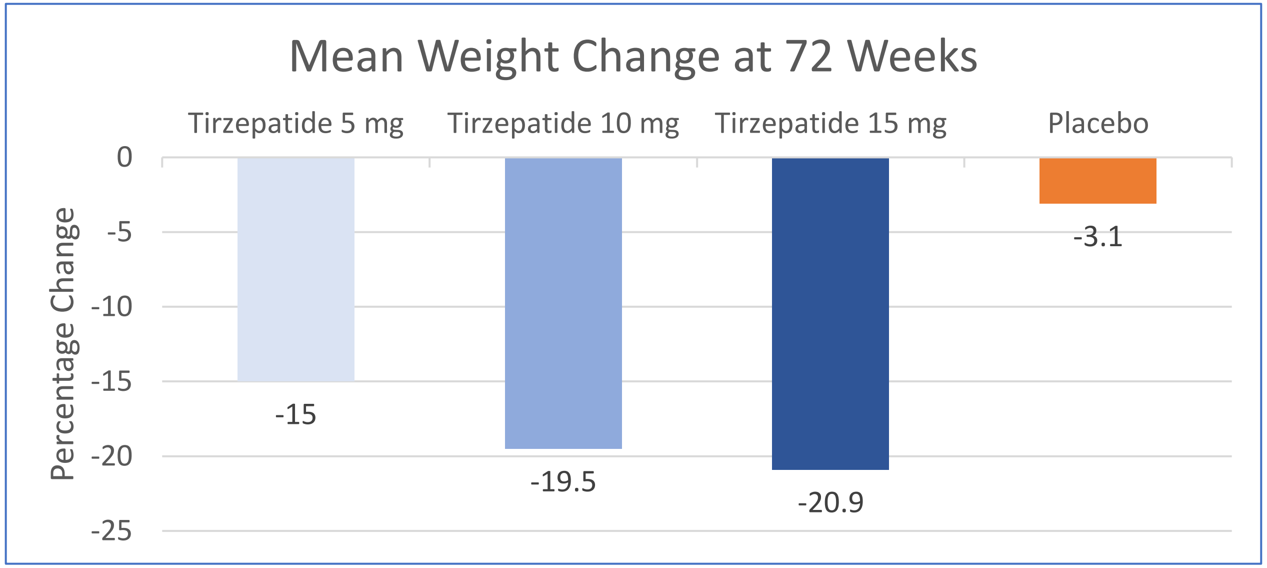
STATISTICAL ANALYSIS

A sample size of 2,400 participants provided greater than 90% power to demonstrate superiority of Tirzepatide over placebo at two-sided significance level of 0.025, assuming at least 11% difference in weight reduction, SD of 10% and 25% dropout rate, with analysis conducted in the ITT population.

RESULTS

COPRIMARY OUTCOMES

Percentage Weight Reduction at 72 Weeks



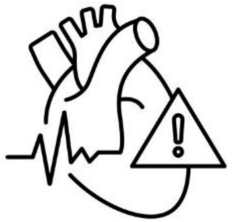
Tirzepatide 5 mg	Tirzepatide 10 mg	Tirzepatide 15 mg	Placebo
-15.0% *mean weight change*	**-19.5%** *mean weight change*	**-20.9%** *mean weight change*	**-3.1%** *mean weight change*
85% achieved ≥5% weight reduction	**89%** achieved ≥5% weight reduction	**91%** achieved ≥5% weight reduction	**35%** achieved ≥5% weight reduction

All three Tirzepatide doses showed statistically significant superiority over Placebo (P<0.001)

RESULTS

SECONDARY OUTCOMES

Significant improvements compared to Placebo



Cardiometabolic Benefits

Waist Circumference ↓14.0-18.5 cm	**Systolic BP** ↓ 7.2 mmHg (pooled Tirzepatide)	**Fasting Insulin** ↓ 42.9%
Triglycerides ↓ 24.8%	**HDL-C** ↑ 8.0%	**Total Fat Body Mass** ↓ 33.9% (3x times greater than lean mass reduction)

Prediabetes Reversal

95.3% of participants with prediabetes at baseline had reverted to normoglycemia with Tirzepatide groups (vs. 61.9% with Placebo group).

Physical Function

SF-36 physical function scores improved more with Tirzepatide, indicating improved participant-reported physical functioning.

SAFETY PROFILE & TOLERABILITY

OVERALL SAFETY

At least one AE:
78.9 to 81.8% (Tirzepatide) vs.72.0% (placebo)
Trial completion rate:
- 86% overall
- Tirzepatide: 88.4 to 89.8%
- Placebo: 77.0%

MOST COMMON AEs

- Gastrointestinal (nausea, diarrhea, and constipation)
- Mild to moderate in severity
- Occurred primarily during the dose-escalation period
- Transient in nature

SERIOUS ADVERSE EVENTS

Overall Incidence: 6.3% (similar across treatment groups)

Note: ~21% of serious adverse events were COVID-19 related (trial conducted during pandemic)

TREATMENT DISCONTINUATIONS DUE TO AEs

- 5 mg: 4.3%
- 10 mg: 7.1%
- 15 mg: 6.2%
- Placebo: 2.6%

NOTABLE SAFETY FINDINGS

❖ Four cases of adjudication-confirmed pancreatitis were reported (evenly distributed, none severe).
❖ No cases of medullary thyroid cancer were reported.

SUMMARY & CLINICAL IMPLICATIONS

Substantial Weight Loss: Tirzepatide achieved mean weight reduction of 15.0-20.9%, substantially exceeding previous anti-obesity medications.

Higher Response Rates: 89-91% of participants achieved ≥5% weight reduction and 50-57% of participants achieved ≥10% weight reduction with higher doses.

Cardiometabolic Benefits: Improvements in BP, insulin sensitivity, lipids, body composition and SF-36 physical function score.

Acceptable Safety Profile: Predominantly mild to moderate gastrointestinal side effects, consistent with incretin-based therapies.

Tirzepatide demonstrates a significant advancement in obesity treatment, with a novel glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1) receptor agonism providing substantial, sustained weight reduction and metabolic improvements that reduce cardiovascular and diabetes risk.

Abbreviations: AE, Adverse Event; BP, Blood Pressure; COVID-19, Coronavirus Disease 2019; DXA, Dual-energy X-ray Absorptiometry; GIP, Glucose-dependent Insulinotropic Polypeptide; GLP-1, Glucagon-like Peptide-1; HDL-C, High-Density Lipoprotein Cholesterol; ITT, Intention-to-Treat; QoL, Quality of Life; SD, Standard Deviation; SF-36, 36-Item Short Form Health Survey

Reference: Jastreboff AM, Aronne LJ, Ahmad NN, et al; SURMOUNT-1 Investigators. Tirzepatide Once Weekly for the Treatment of Obesity. N Engl J Med. 2022 Jul 21;387(3):205-216. doi: 10.1056/NEJMoa2206038. Epub 2022 Jun 4. PMID: 35658024.